

registers an image of whatever has pressed down on the floor.

Batteries to become more powerful

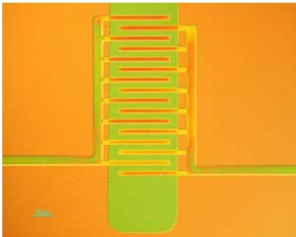
A technology that could lead to more powerful, lightweight batteries is being developed by a team of researchers at Massachusetts Institute of Technology (MIT), USA. Yang Shao-Horn, associate professor of mechanical engineering, MIT, says that many groups have been pursuing work on lithium-air batteries—a technology that could advance work on energy density.

Lithium-air batteries work similar to the lithium-ion batteries that are used in portable electronics and are a leading contender for electric vehicles. But because lithium-air batteries replace the heavy conventional compounds in such batteries with a carbon-based air electrode and flow of air, the batteries themselves can be much lighter.

Yang, along with some of her students and visiting professor Hubert Gasteiger, reported that electrodes with gold or platinum as a catalyst show a much higher level of activity and thus a higher efficiency than simple carbon electrodes in these batteries.

Plastics can slash solar panel costs

Replacing indium-tin oxide (ITO) with power-conducting plastics as



In the plastic transistor pictured here, the plastic is moulded into interdigitated electrodes (orange) allowing current flow to and from the active channel (green). Image courtesy: Loo Research Group

the primary conducting material and manufacturing plastic solar cells using low-cost printing techniques hold potential for lowering the cost of solar panels.

Use of polymers for electronics, also known as 'organic electronics,' has so far been hampered by the loss of conductivity associated with mouldable plastics. Yueh-Lin Loo, associate professor of chemical engineering, and his team at Princeton University have found that in making the polymers mouldable, their structures are trapped in a rigid form, which prevented electrical current from travelling through them. So they developed a way to relax the struc-

ture of the plastics by treating them with an acid after they were processed into the desired form. Using the method, they were able to make a plastic transistor—a fundamental component of electronics that is used to amplify and switch electronic signals.

They produced the electrodes of the transistor by printing the plastic onto a surface—a fast and cheap method similar to the

way an inkjet printer produces a pattern on a piece of paper.

Organic semiconductor for solar cells and displays

Organic semiconductor may become a viable candidate for creating large-area electronics such as solar cells and displays, claims National Institute of Standards and Technology (NIST). It can be sprayed onto a surface as easily as paint through NIST's airbrush technique that deposits a material to create spray-on transistors.

"At this stage, there is no established best material or manufacturing process for creating low-cost, large-area electronics. What our team

Wide portfolio, broad spectrum – your partner in T&M

Audio analyzers

Universal instruments flexibly configurable for measurements on analog and digital interfaces

Signal generators

Covering all instrument classes for digital, analog, baseband, RF and microwave applications

Signal and spectrum analyzers

Largest product portfolio on the market from handheld to mid-range to the fastest and most exacting analyzers available



Email: sales@rsindia.rohde-schwarz.com

Visit: www.rohde-schwarz.co.in

NEW DELHI: Tel: 011-42535400, Fax: 011-42535433; BANGALORE: Tel: 080-41780400, Fax No: 080-41780444